



Dennis MacDonald/Photoedit

Gambler's fallacy can cause people to make biased decisions.

People expect that a small sample of roulette wheel spins (such as 8) will show the same proportion of reds as will a very large sample (such as 100,000). However, the chances of finding large deviations from the expected proportion are much greater with a small N sample. Said another way, only very large samples can be expected to be representative of the population from which they come. Sedlmeier and Gigerenzer (2000) explored the issue of people's intuitions about sample size in greater depth, arguing that people sometimes do have correct intuitions about sample size, but often don't.

"Man who" arguments are another example of the misuse of the representativeness heuristic. The term was coined by Nisbett and Ross (1980). A "man who" argument is usually advanced by someone who has just confronted, for instance, a statistical summary of a number of cases reporting that lung cancer rates are significantly higher among smokers than nonsmokers. The reply "I know a *man who* smoked three packs a day and lived to be 110" is a particularly vivid example of ignoring base rate information and instead paying as much attention to small sample sizes (the individual man who was known, $N = 1$) as to large ones (those cases summarized, where N may be 10,000 or more).

Framing Effects

Driving down the road, you notice your car is running low on gasoline, and you see two service stations, both advertising gasoline. Station A's price is \$2.00 per

gallon; station B's, \$1.95. Station A's sign also announces, "5 cents/gallon discount for cash!" Station B's sign announces, "5 cents/gallon surcharge for credit cards." All other factors being equal (for example, cleanliness of the stations, whether you like the brand of gasoline carried, number of cars waiting at each), to which station would you choose to go? Many people report a preference for Station A, the one that offers a cash discount (Thaler, 1980).

It is interesting that people have this preference, because both stations are actually offering the same deal: a price of \$1.95 per gallon if you use cash and \$2.00 per gallon if you use a credit card. Tversky and Kahneman (1981) explained this phenomenon in terms of **framing effects**: People evaluate outcomes as changes from a reference point, their current state. Depending on how their current state is described, they perceive certain outcomes as gains or losses. The description is therefore said to "frame" the decision, or to provide a certain context for it. We have already seen with previous cognitive topics (such as perception, thinking, reasoning) that context effects can play a large role in affecting cognitive performance. Framing effects, in essence, can be thought of as context effects in decision making.

Here's what appears to be going on in the gas station example. Described as a "cash discount," the price seems a bargain—you assume you are starting from a reference point of \$2.00 a gallon and then saving or gaining a nickel. In the case of station B, however, you describe the situation to yourself as follows: "OK, so they're charging \$1.95. Sounds good. But hey, wait a minute. If I want to use my card, they'll jack up the price to \$2.00. Hey, I'd lose a nickel a gallon that way. What rip-off artists they are! Heck, I'll just go to station A." Kahneman and Tversky (1979) argued that we treat losses more seriously than we treat gains of an equivalent amount (whether of money or of some other measure of satisfaction). That is, we care more about losing a dollar than we do about gaining a dollar, or more about losing a nickel than gaining a nickel.

The problem is that simply changing the *description* of a situation can lead us to adopt different reference points and therefore to see the same outcome as a gain in one situation and a loss in the other. That in turn may lead us to change our decision, not because anything in the problem has changed but simply because the way we describe the situation to ourselves has changed.

Anchoring

Suppose I ask you to answer a numerical question with an estimate (assuming you don't know the exactly correct value): As of April 2000, what was the population of Philadelphia? (I'll give you the correct answer later.) Imagine I give two people, call them Tim and Kim, this question, but I give each one a "starting value," obtained by spinning a roulette wheel. Now, Kim and Tim watch me spin the wheel, and they know that it operates (and stops) purely by chance

and that the “starting value” is arbitrary. Kim’s starting value is 1 million; Tim’s, 2 million. If they are like most research participants in Tversky and Kahneman’s (1973) study, Kim will arrive at an estimate of 1.25 million; Tim, 1.75 million. In other words, their initial starting point will have a huge effect on their final estimates, showing evidence of the phenomenon known as **anchoring**. (The correct value is 1,517,550, according to the April 1, 2000, U.S. Census.)

Likewise, consider two groups of high school students, each given five seconds to estimate a complex expression. Group 1 estimates $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$, and reports a mean estimate of 2250; Group 2 estimates $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8$, and reports the answer to be (on average) 512. As you can tell, both problems are identical. You probably are not able to tell this quickly, but both estimates are too small: The correct value is 40,320.

Tversky and Kahneman (2000) explain these results this way: People tend to perform the first few steps of multiplication, then extrapolate. The extrapolation tends to be too little rather than too much. That explains the fact that both groups of participants underestimated the answer. In addition, those who started with $1 \times 2 \times 3$ began with a smaller value than did those who began with $8 \times 7 \times 6$, so the first group more severely underestimated the result.

Sunk Cost Effects

A major educational initiative is begun in your hometown; \$3 million is invested to help students stay away from cigarettes, liquor, and other drugs. In the third of four years, evidence begins to accumulate that the program is not working. A local legislator proposes ending funding to the program before the scheduled date. Howls of protest go up from some individuals, who claim that to stop a program after a large expenditure of funds has been spent would be a waste. These individuals are falling prey to what Arkes and Blumer (1985) have dubbed the **sunk cost effect**: “[the] greater tendency to continue an endeavor once an investment in money, effort, or time has been made” (p. 124).

Why is this an error? The explanation goes something like this: Money spent is already gone. Whether or not a great deal of money (or time, or energy, or emotion) has been spent does not affect the likelihood of future success. Those resources have been used, regardless of which option is chosen. All that should affect a decision, therefore, are the expected future benefits and costs of each option (Arkes & Hutzler, 2000).

Illusory Correlation

You and a friend, both students of (if not majors in) psychology, observe fellow students around campus and discover a behavioral pattern you call “hair twisting”: The person pinches a strand of hair between thumb and forefinger and